上节回顾

间隔. $ds^2 = (c\Delta t)^2 - |\Delta \vec{x}|^2$
$$\begin{cases} |c\Delta t| = |\Delta\vec{x}| & \text{类光间隔（由光信号联系）} \\ |c\Delta t| > |\Delta\vec{x}| & \text{类时间隔（满足因果律）} \\ |c\Delta t| < |\Delta\vec{x}| & \text{类空间隔} \end{cases}$$

物理量按时空变换性质分类

标量. ds, $d\tau$ $\quad \frac{\partial}{\partial x_\mu}\frac{\partial}{\partial x_\mu} = \Box = \nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}$

矢量： $x_\mu' = \alpha_{\mu\nu} x_\nu$

$V_\mu' = \alpha_{\mu\nu} V_\nu$

例. $\frac{\partial}{\partial x_\mu} = (\nabla, \frac{\partial}{\partial(ict)})$, $\quad U_\mu = \frac{dx_\mu}{d\tau} = \gamma(\vec{u}, ic)$, $\quad J_\mu = \rho_0 U_\mu = (\vec{J}, ic\rho)$

$k_\mu = (\vec{k}, i\frac{\omega}{c})$ $\quad A_\mu = (\vec{A}, i\frac{\varphi}{c})$

二阶张量: $D_{ij}' = \alpha_{im}\alpha_{jn} D_{mn}$

物理规律的协变性

电荷守恒定律： $\nabla\cdot\vec{J} + \frac{\partial\rho}{\partial t} = 0$ $\quad (\nabla, \frac{\partial}{\partial ict})\begin{pmatrix}\vec{J} \\ ic\rho\end{pmatrix} = 0$ $\quad \partial_\nu J_\nu = 0$

达朗贝尔方程： $\Box A_\mu = -\mu_0 J_\mu$ $\quad$ 洛伦兹条件： $\partial_\mu A_\mu = 0$

$A_\mu' = \alpha_{\mu\nu} A_\nu$

四维反对称张量: $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu = \begin{pmatrix} 0 & B_3 & -B_2 & \frac{E_1}{ic} \\ -B_3 & 0 & B_1 & \frac{E_2}{ic} \\ B_2 & -B_1 & 0 & \frac{E_3}{ic} \\ -\frac{E_1}{ic} & -\frac{E_2}{ic} & -\frac{E_3}{ic} & 0 \end{pmatrix}$

$$\begin{cases} \nabla\cdot\vec{E} = \frac{\rho}{\varepsilon_0} \\ \nabla\times\vec{B} = \mu_0\vec{J} + \mu_0\varepsilon_0\frac{\partial\vec{E}}{\partial t} \end{cases} \Rightarrow \partial_\nu F_{\mu\nu} = \mu_0 J_\mu$$

如

$$\partial_\nu F_{4\nu} = \partial_1 F_{41} + \partial_2 F_{42} + \partial_3 F_{43} + \partial_4 F_{44}$$
$$= \partial_x(-\frac{E_1}{ic}) + \partial_y(-\frac{E_2}{ic}) + \partial_z(-\frac{E_3}{ic})$$
$$= \frac{i}{c}\nabla\cdot\vec{E} = \mu_0 J_4 = \mu_0 ic\rho$$
$$\nabla\cdot\vec{E} = \mu_0 c^2\rho = \mu_0\frac{1}{\mu_0\varepsilon_0}\rho = \rho/\varepsilon_0$$

$$\partial_\nu F_{1\nu} = \partial_1 F_{11} + \partial_2 F_{12} + \partial_3 F_{13} + \partial_4 F_{14}$$
$$= \partial_y B_3 - \partial_z B_2 + \frac{\partial}{\partial(ict)}\frac{E_1}{ic}$$
$$= (\nabla\times\vec{B})_1 - \frac{1}{c^2}\frac{\partial E_1}{\partial t} = \mu_0 J_1$$

$$\begin{cases}\nabla\cdot\vec{B}=0\\ \nabla\times\vec{E}=-\frac{\partial\vec{B}}{\partial t}\end{cases}\Rightarrow \partial_\lambda F_{\mu\nu}+\partial_\mu F_{\nu\lambda}+\partial_\nu F_{\lambda\mu}=0$$

如：$\mu=1,\ \nu=2,\ \lambda=3$

$$\partial_3 F_{12}+\partial_1 F_{23}+\partial_2 F_{31}=0$$

$$\partial_z B_3+\partial_x B_1+\partial_y B_2=0$$

$$\nabla\cdot\vec{B}=0$$

$\mu=4,\ \nu=1,\ \lambda=2$

$$\partial_2 F_{41}+\partial_4 F_{12}+\partial_1 F_{24}=0$$

$$\partial_y\left(-\frac{E_1}{ic}\right)+\frac{\partial}{\partial ict}B_3+\partial_x\frac{E_2}{ic}=0$$

$$-\frac{i}{c}(\nabla\times\vec{E})_3-\frac{i}{c}\left(\frac{\partial\vec{B}}{\partial t}\right)_3=0$$

$$(\nabla\times\vec{E})_3=-\left(\frac{\partial\vec{B}}{\partial t}\right)_3$$

也可以引入另一张量 $G_{\mu\nu}$（由 $F_{\mu\nu}$ 通过替换 $\vec{E}/c\to\vec{B}$，$\vec{B}\to-\vec{E}/c$ 得到），则

$$G_{\mu\nu}=\begin{pmatrix}0 & -E_3/c & E_2/c & B_1/i\\ E_3/c & 0 & -E_1/c & B_2/i\\ -E_2/c & E_1/c & 0 & B_3/i\\ -B_1/i & -B_2/i & -B_3/i & 0\end{pmatrix}$$

$$\partial_\nu G_{\mu\nu}=0$$

如 $\mu=4$

$$\partial_1 G_{41}+\partial_2 G_{42}+\partial_3 G_{43}+\partial_4 G_{44}=0$$

$$i(\partial_1 B_1+\partial_2 B_2+\partial_3 B_3)=0$$

$$\nabla\cdot\vec{B}=0$$

$\mu=1$

$$\partial_1 G_{11}+\partial_2 G_{12}+\partial_3 G_{13}+\partial_4 G_{14}=0$$

$$-\partial_2(E_3/c)+\partial_3(E_2/c)-i\frac{\partial}{\partial ict}B_1=0$$

$$-(\nabla\times\vec{E})_1-\left(\frac{\partial\vec{B}}{\partial t}\right)_1=0$$

$$(\nabla\times\vec{E})_1=-\left(\frac{\partial\vec{B}}{\partial t}\right)_1$$

由张量变换关系

$$F'_{\mu\nu}=\alpha_{\mu m}\alpha_{\nu n}F_{mn}$$

导出电磁场的变换关系、

$$E_1'=E_1 \qquad B_1'=B_1$$

$$E_2'=\gamma(E_2-vB_3) \qquad B_2'=\gamma\left(B_2+\frac{v}{c^2}E_3\right)$$

$$E_3'=\gamma(E_3+vB_2) \qquad B_3'=\gamma\left(B_3-\frac{v}{c^2}E_2\right)$$

或用矢量形式写为

$\vec{E}_\parallel' = \vec{E}_\parallel$ $\qquad\qquad$ $\vec{B}_\parallel' = \vec{B}_\parallel$

$\vec{E}_\perp' = \gamma(\vec{E}+\vec{v}\times\vec{B})_\perp$ $\qquad\qquad$ $\vec{B}_\perp' = \gamma(\vec{B}-\frac{\vec{v}}{c^2}\times\vec{E})_\perp$

动生电动势的物理解释

例. 求匀速运动的点电荷的场

解. 设S'系原点固定在点电荷上.

$$\vec{E}'(\vec{x}',t) = \frac{q\vec{x}'}{4\pi\varepsilon_0|\vec{x}'|^3} \qquad \vec{B}'(\vec{x}',t)=0$$

S系相对于S'系沿x轴以速度$-v$运动

$$\begin{cases} E_x = E_x' \\ E_y = \gamma(E_y'+vB_z') = \gamma E_y' \\ E_z = \gamma(E_z'-vB_y') = \gamma E_z' \\ B_x = B_x' = 0 \\ B_y = \gamma(B_y' - \frac{v}{c^2}E_z') = -\gamma\frac{v}{c^2}E_z' = -\frac{v}{c^2}E_z \\ B_z = \gamma(B_z' + \frac{v}{c^2}E_y') = \gamma\frac{v}{c^2}E_y' = \frac{v}{c^2}E_y \end{cases}$$

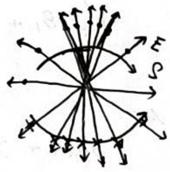
设$t=0$时刻，点电荷正好与S系的原点重合，并且我们在这一时刻测量空间的场. 于是

$$x'=\gamma x \qquad y'=y \qquad z'=z \qquad t'=-\gamma\frac{v}{c^2}x$$

$$|\vec{x}'| = \sqrt{x'^2+y'^2+z'^2} = \sqrt{(\gamma x)^2+y^2+z^2}$$

$$\vec{E} = \frac{\gamma q\vec{x}}{4\pi\varepsilon_0[(\gamma x)^2+y^2+z^2]^{3/2}}$$

$$\vec{B} = \frac{\vec{v}}{c^2}\times\vec{E}$$



讨论：1) 电场仍沿径向，但强度分布不再是球对称. 运动方向的场很微弱

2) 磁流在以电荷为中心的球面上流动. 伴随着电荷的运动作定向流